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CHE 321 — Past Questions & Materials
by Renance

Course: CHE 321
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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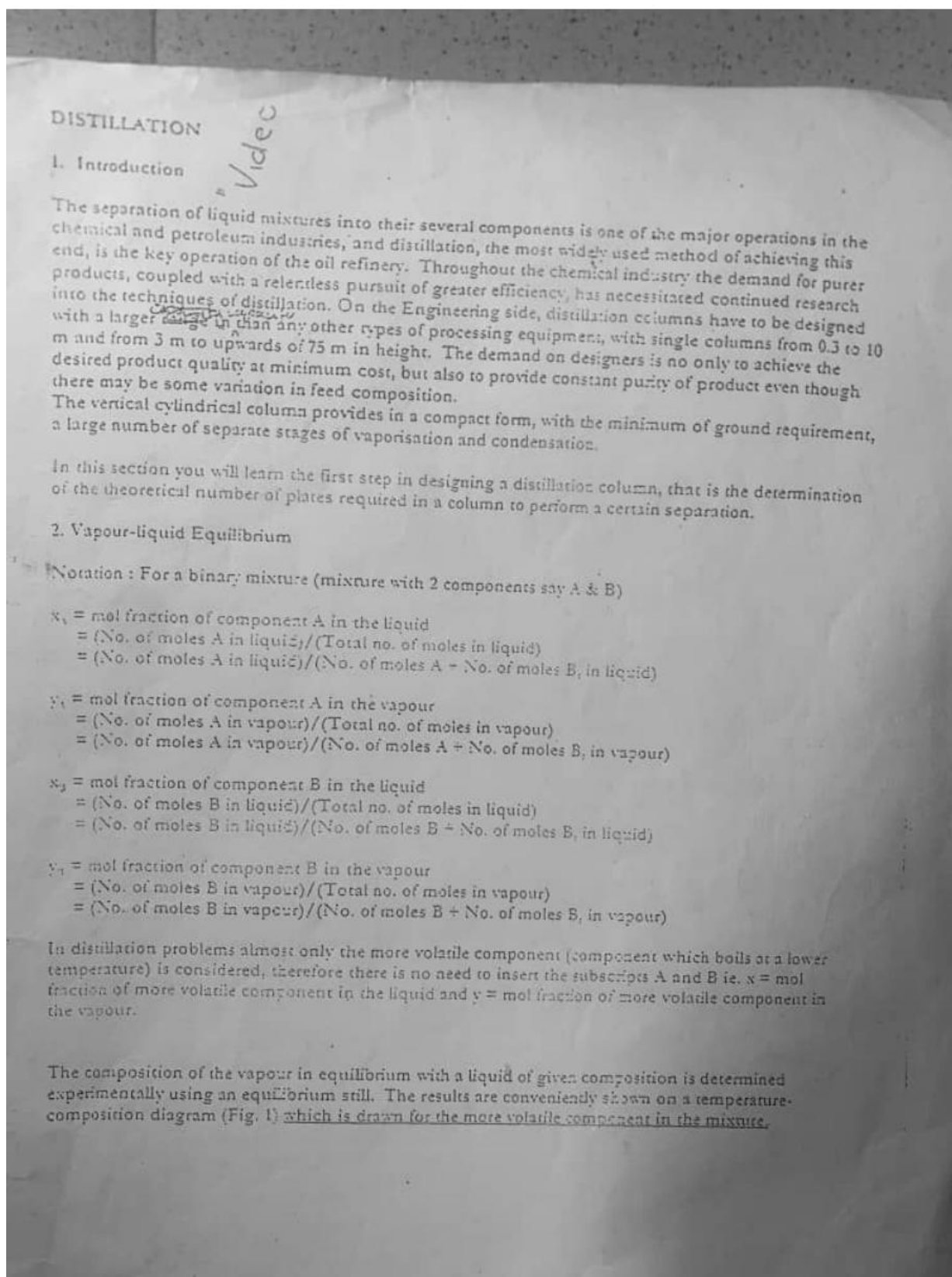
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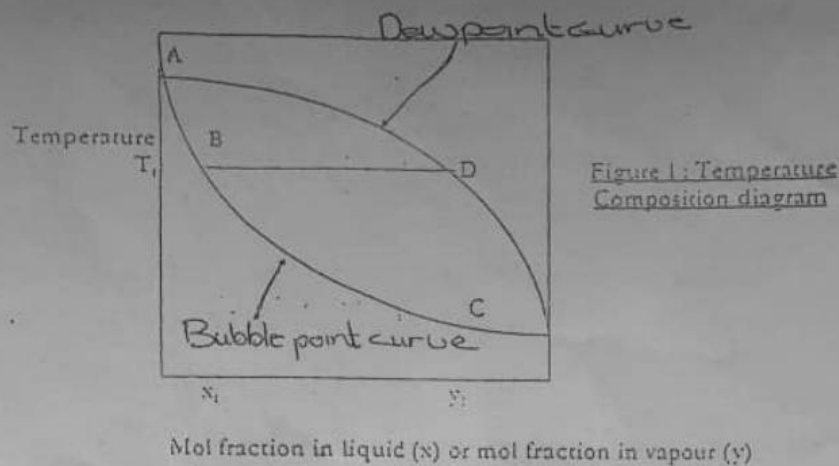
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Paper 1 of 1 — CHE 321

Level 300 • Second Semester • Handout

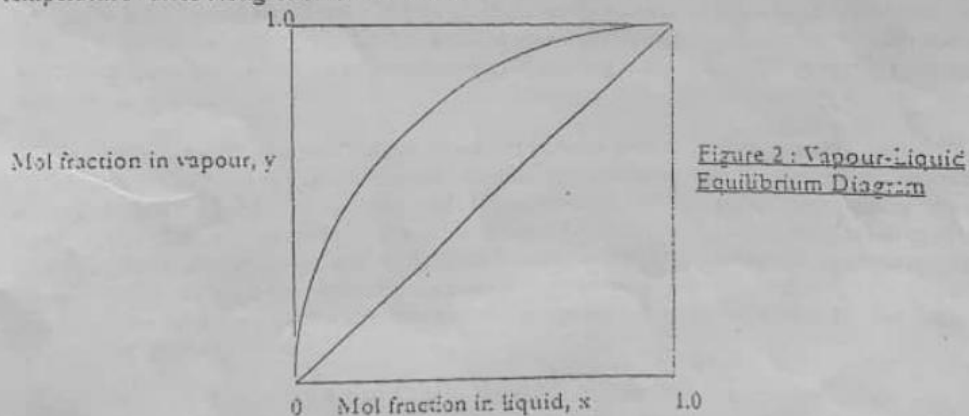
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The curve ABC shows the composition of the liquid which boils at any given temperature, and the curve ADC the corresponding composition of the vapour at that temperature. Thus, a liquid of composition x_1 will boil at temperature T_1 and the vapour in equilibrium is indicated by point D of composition y_1 . It is seen that for any liquid composition x the vapour formed will be richer in the more volatile component (where x is the mol fraction of the more volatile component in the liquid, and y in the vapour).

For distillation purposes it is more convenient to plot y against x , for the more volatile component in the mixture, at a constant pressure, since the majority of industrial distillations take place at substantially constant pressure. This is shown in Fig. 2 where it must be remembered that the temperature varies along each of the curves.



3. Partial Vaporisation and Partial Condensation

Suppose a mixture of benzene and toluene is heated in a vessel, closed in such a way that the pressure remains atmospheric but no material can escape. If the mol fraction of the more volatile component (benzene) in the liquid is plotted as abscissa, and the temperature at which the mixture boils as ordinate, then the boiling curve is obtained as shown by $ABCJ$ (Figure 3). The corresponding dew point curve $ADEJ$ shows the temperature at which a vapour of composition y starts to condense.

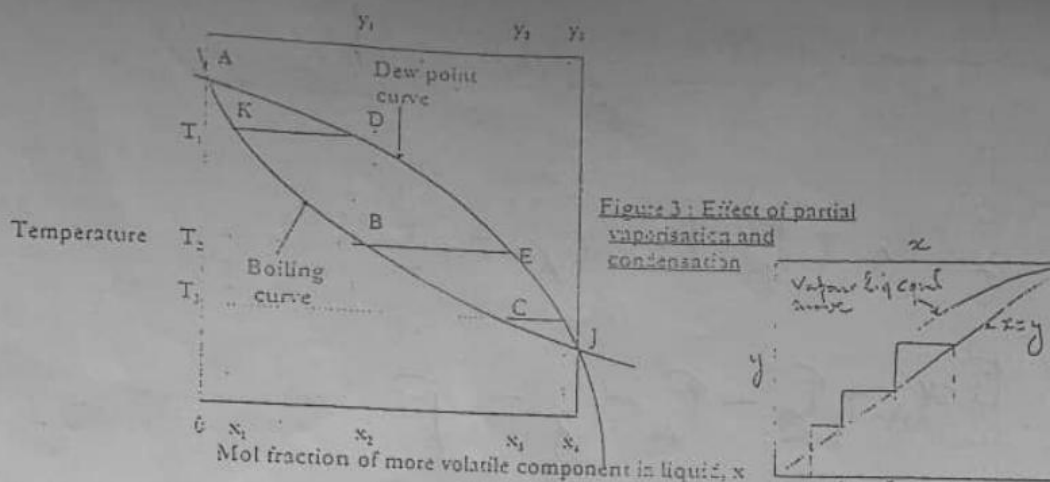


Figure 3: Effect of partial vaporisation and condensation

If a mixture of composition x_1 which is at a temperature below its boiling point, is heated at constant pressure it will begin to boil at temperature T_1 as shown by point K on the diagram. Thus vapour with composition y_1 is formed. If this vapour is then condensed to a point below B, the vapour is converted to liquid with composition x_2 which is equal to y_1 . If this liquid x_2 is again heated to point B it will boil and form vapour with composition y_2 . If this vapour is again condensed and reboiled vapour of composition y_3 is formed.

It can be seen that after every condensing and reboiling step, the composition of the more volatile component in the vapour, y increases. Thus if these steps are repeated sufficiently, the final vapour formed will be almost entirely made up of the more volatile component and the remaining liquid almost entirely made up of the less volatile component. In this way the more volatile component can be separated from the less volatile component by removing it as vapour and removing the less volatile component as liquid.

A distillation column operates on the same principle as above. The column has plates or trays within it. The liquid is boiled at the bottom of the column, the resulting vapour rises to the first plate from the bottom where it partially condenses and is then reboiled, producing vapour richer in the more volatile component. This vapour rises to the 2nd plate where it is condensed and reboiled producing vapour even richer in the more volatile component. This process is repeated on the remaining plates until a vapour almost pure with the more volatile component is removed as the top product at the top of the column and a liquid almost pure with the less volatile component is removed as the bottom product or residue at the bottom of the column.

Each step in the Figure 3 denotes a stage in the distillation column, so by counting the steps in the diagram, we can determine the theoretical no of stages required to perform a certain separation.

3. Partial Pressures, and Dalton's, Raoult's, and Henry's Laws

The abovementioned laws govern vapour-liquid equilibria and can be used to draw equilibrium and composition diagrams.

The partial pressure P_A of component A in a mixture of vapours is the pressure that would be exerted by component A at the same temperature, if present in the same volumetric concentration as in the mixture.

By Dalton's Law of partial pressures, $P = \sum P_i$ that is the total pressure is equal to the summation of the partial pressures. Then, since in an ideal gas or vapour the partial pressure is proportional to the mol fraction of the constituent:

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20

$$P_A = y_A P \dots\dots\dots(1)$$

For an *ideal mixture*, the partial pressure is related to the concentration in the liquid phase by Raoult's Law which can be written:

$$P_A = x_A P_A^* \dots\dots\dots(2)$$

where P_A^* is the vapour pressure of pure A at the same temperature. This relation is usually found to be true only for high values of x_A or correspondingly low values of x_B but mixtures of organic isomers and some hydrocarbons follow the law closely. For low values of x_A , a linear relation between P_A and x_A again exists, but the proportionality factor is Henry's constant H , and not the vapour pressure P^* of the pure material.

For a liquid solute A in a solvent liquid B, Henry's Law takes the form:

$$P_A = H x_A \dots\dots\dots(3)$$

If it is known that the mixture follows Raoult's Law, then the values of y_A , for various values of x_A may be calculated from a knowledge of the vapour pressures of the two components at various temperatures. Thus,

$$\begin{aligned} P_A &= P_A^* x_A \\ P_A &= P y_A \end{aligned}$$

and

so

$$y_A = P_A^* x_A / P \text{ and } y_B = P_B^* x_B / P \dots\dots\dots(4)$$

but

$$y_A + y_B = 1 \text{ and } x_A + x_B = 1 \dots\dots\dots(5)$$

$$P_A^* x_A / P + P_B^* (1 - x_A) / P = 1 \dots\dots\dots(6)$$

giving

$$x_A = (P - P_B^*) / (P_A^* - P_B^*) \dots\dots\dots(7)$$

Do exercise (a)

4. The Fractionating Process

The column consists of a cylindrical structure divided into sections by a series of perforated trays which permit the upward flow of vapour. The liquid reflux flows across each tray, over a weir, down a downcomer to the tray below. The vapour rising from the top tray passes to a condenser and then via an accumulator or reflux drum and a reflux divider, where part is withdrawn as the overhead product D, and the remainder is returned to the top tray as reflux R.

The liquid in the base of the column is frequently heated, either by condensing steam or by a hot oil stream, and the vapour rises through the perforations to the bottom tray. A more commonly used arrangement is with an external reboiler where the liquid from the still passes into the reboiler where it flows over the tubes and weir and leaves as the bottom product via a bottoms cooler, which preheats the incoming feed. The vapour generated in the reboiler is returned to the bottom of the column with a composition y_1 and enters the bottom tray where it is partially condensed and then re-vaporised to give vapour of composition y_1 . This operation of partial condensation of the rising vapour and partial vaporisation of the reflux liquid is repeated on each tray. Vapour of composition y_1 from the top tray is condensed to give the top product D and the reflux R, both of the same composition y_1 . The feed stream is introduced on some intermediate tray where the liquid has approximately the same composition as the feed. The part of the column above the feed point is known as the rectifying section; the lower portion is known as the stripping section. On an ideal tray the vapour rising from it will be in equilibrium with the liquid leaving, though in practice a smaller degree of enrichment will occur.

In analysing the operation on each tray it is important to note that the vapour rising to it, and the reflux flowing down to it, are not in equilibrium, and adequate rates of mass and heat transfer are essential for the proper functioning of the tray.

The tray described above is known as a sieve tray and has perforations of up to about 12 mm diameter, but there are several alternative arrangements for promoting mass transfer on the tray eg. valve units, bubble caps and other devices. In all cases the aim is to promote good mixing of vapour and liquid with a low drop in pressure across the tray.

On each tray the system tends to reach equilibrium because:

- (a) Some of the less volatile component condenses from the rising vapour into the liquid thus increasing the concentration of the more volatile component (MVC) in the vapour.
- (b) Some of the MVC is vaporised from the liquid on the tray thus decreasing the concentration of the MVC in the liquid.

The number of molecules passing in each direction from vapour to liquid and in reverse will be approximately the same since the heat given out by one mol of the vapour on condensing is approximately equal to the heat required to vaporise one mol of the liquid. The problem is thus one of equimolecular counterdiffusion. If the molar heats of vaporisation are approximately constant, the flows of liquid and vapour in each part of the column will not vary from tray to tray. This is the concept of constant molar overflow.

5. Number of Plates Required in a Distillation Column

5.1. Notation

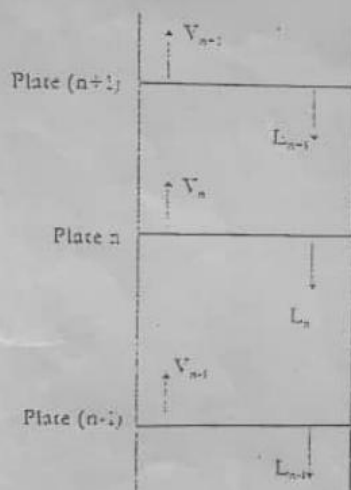


Figure 4: Material balance over a plate

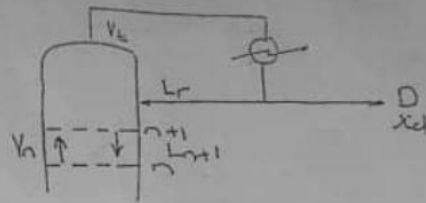
Plate n receives liquid L_{n+1} from plate $n + 1$ above, and vapour V_{n+1} from plate $n - 1$ below. Plate n supplies liquid L_n to plate $n - 1$, and vapour V_n to plate $n + 1$.

The subscript n denotes any plate in the enriching or rectifying section of the column. For the stripping section the subscript m is used.

The action of the plate is to bring about mixing so that the vapour V_n of composition y_n approaches equilibrium with the liquid L_n of composition x_n . The streams L_{n+1} and V_{n+1} cannot be in equilibrium and, during the interchange process on the plate, some of the more volatile component is vaporised from the liquid L_{n+1} decreasing its concentration to x_n and some of the less volatile component is condensed from V_{n+1} increasing the vapour concentration to y_n . The heat to vaporise the more volatile component is supplied by partial condensation of the vapour V_{n+1} . Thus the resulting vapour rising up, whilst the less volatile component is transferred in the opposite direction to the

balances on Distillation Column

up:



(n+1)

$$V_n + L_r = L_{n+1} + V_{n+1}$$

$$V_n = L_{n+1} + D$$

$$V_n + L_r = L_{n+1} + V_{n+1} + D$$

$$V_n = L_{n+1} + D$$

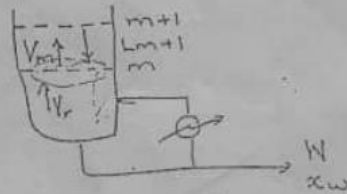
$$V_n \cdot y_n = L_{n+1} \cdot x_{n+1} + D \cdot x_D$$

$$y_n = \frac{L_{n+1}}{V_n} \cdot x_{n+1} + \frac{D}{V_n} \cdot x_D$$

$$\text{but } L_{n+1} = L_n$$

$$y_n = \frac{L_n}{V_n} \cdot x_{n+1} + \frac{D}{V_n} \cdot x_D$$

3. Set:



$$V_m + L_m = L_{m+1} + V_r$$

$$\text{But } V_r = L_m - W$$

$$V_m + L_m = L_{m+1} + L_m - W$$

$$V_m = L_{m+1} - W$$

$$V_m \cdot y_m = L_{m+1} \cdot x_{m+1} - W \cdot x_W$$

$$L_{m+1} = L_m$$

$$y_m = \frac{L_m}{V_m} \cdot x_{m+1} - \frac{W}{V_m} \cdot x_W$$

(32)

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3.2. Calculation of Number of Plates Using the McCabe-Thiele Method

Suppose a unit is operating as shown in Figure 5 so that a binary feed F is distilled to give a top product D and a bottom product W , with x_d , x_n and x_w as the corresponding mol fractions of the more volatile component. The vapour V_1 rising from the top plate is condensed, and part is run back as liquid at its boiling point to the column as reflux, the remainder being withdrawn as product.

6.2.1 Overall Material Balance

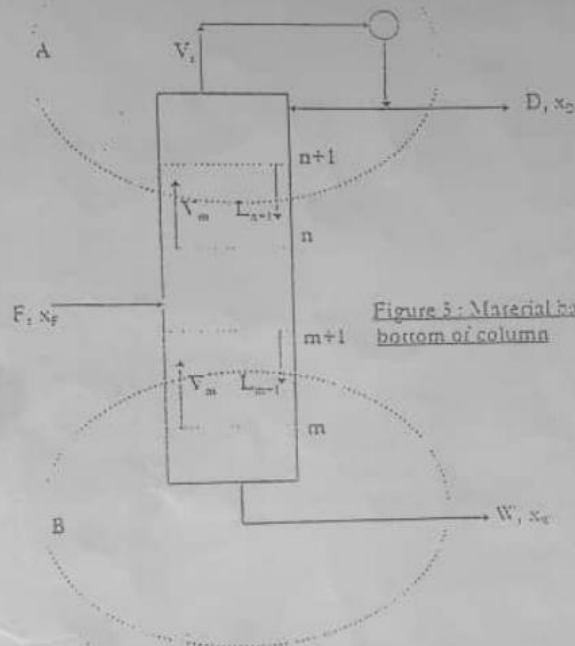


Figure 5: Material balance at top and bottom of column

A material balance above plate n (indicated by the loop A in Figure 5) gives:

$$V_n = L_{n+1} + D \quad \dots\dots\dots(5)$$

Expressing this balance on the more volatile component:

$$y_n V_n = L_{n+1} x_{n+1} + D x_d \quad \dots\dots\dots(9)$$

Thus

$$y_n = \frac{L_{n+1}}{V_n} x_{n+1} + \frac{D}{V_n} x_d \quad \dots\dots\dots(10)$$

Since the molar liquid overflow is constant, L_n may be replaced by L_{n+1} and

$$y_n = \frac{L_n}{V_n} x_{n+1} + \frac{D}{V_n} x_d \quad \dots\dots\dots(11)$$

This equation gives a relation between the composition of the vapour rising to the plate and the composition of the liquid on any plate above the feed plate (enriching section). Equation (11) governs the relationship between the liquid and vapour in the enriching section of the column and is the equation of the enriching section, or top, operating line.

n : number of stages shown on the diagram.
 therefore for the example in figure 6, the no. of theoretical stages required = 3 and the
 neoretical no. of plates or trays required = 3 - 1 = 2 ie. theoretical no. of plates or trays required = no.
 of theoretical stages required minus one.

5.3.3. The Intersection of the Operating Lines

The point of intersection of any 2 lines is a point common to both lines. If the 2 operating lines intersect at a point (x_3, y_3) then since this point is common to both operating lines it must satisfy both the equations of the operating lines ie. equations (9) & (13). Therefore

$$y_3 V_m = L_{m+1} x_3 + D x_D \quad \dots\dots\dots (15)$$

and

$$y_3 V_m = L_m x_3 - W x_W \quad \dots\dots\dots (16)$$

Therefore

$$y_3 (V_m - V_n) = (L_m - L_n) x_3 - (D x_D + W x_W) \quad \dots\dots\dots (17)$$

A material balance over the feed plate gives

$$F + L_n + V_m = L_{m+1} + V_n$$

or

$$V_m - V_n = L_{m+1} - L_n - F \quad \dots\dots\dots (18)$$

NR

To obtain a relation between L_n and L_{m+1} it is necessary to make an enthalpy balance over the feed plate, and to consider what happens when the feed enters the column. If the feed is all in the form of liquid at its boiling point, the reflux L_n overflowing to the plate below will be $L_n + F$. If however the feed is a liquid at a temperature T_f less than the boiling point, some vapour rising from the plate below will condense to provide sufficient heat to bring the feed liquor to the boiling point.

Let H_F be the enthalpy per mol of feed, and H_{FG} the enthalpy of one mol of feed at its boiling point. Then the heat to be supplied to bring feed to the boiling point is $F(H_{FG} - H_F)$ and the number of mols of vapour to be condensed to provide this heat is $F(H_{FG} - H_F)/\lambda$, where λ is the molar latent heat of the vapour.

Then, reflux liquor

$$L_{m+1} = L_n + F + F(H_{FG} - H_F)/\lambda,$$

$$= L_n + F(\lambda + H_{FG} - H_F)/\lambda$$

$$L_{m+1} = L_n + qF \quad \dots\dots\dots (19)$$

$$q = \frac{\lambda + H_{FG} - H_F}{\lambda}$$

where q = (heat to vaporise 1 mol of feed) / (molar latent heat of the feed)

Thus from equation (18)

$$V_m - V_n = qF - F \quad \dots\dots\dots (20)$$

A material balance of the more volatile component over the whole column gives

$$F x_F = D x_D + W x_W \quad \dots\dots\dots (21)$$

Thus combining equations (17), (19) & (20) we get

$$F(q-1)y_3 = qF x_3 - F x_F$$

or

$$y_3 = \frac{q}{q-1} x_3 - \frac{x_F}{q-1} \quad \dots\dots\dots (22)$$

This equation is commonly known as the equation of the q -line. If $x_3 = x_F$, then $y_3 = x_F$. Thus, the point of intersection of the two operating lines lies on the straight line of slope $q/(q-1)$ passing

(38)

∴ number of stages shown on the diagram.
 ∴ therefore for the example in figure 6, the no. of theoretical stages required = 3 and the
 theoretical no. of plates or trays required = 3 - 1 = 2 i.e. theoretical no. of plates or trays required = no.
 of theoretical stages required minus one.

3.3.3. The Intersection of the Operating Lines

The point of intersection of any 2 lines is a point common to both lines. If the 2 operating lines intersect at a point (x_4, y_4) then since this point is common to both operating lines it must satisfy both the equations of the operating lines i.e. equations (9) & (13). Therefore

$$y_4 V_n = L_{n+1} x_4 + D x_D \quad \dots\dots\dots(15)$$

and

$$y_4 V_m = L_n x_4 - W x_W \quad \dots\dots\dots(16)$$

Therefore

$$y_4 (V_n - V_m) = (L_m - L_n) x_4 - (D x_D + W x_W) \quad \dots\dots\dots(17)$$

A material balance over the feed plate gives

$$F + L_n + V_m = L_{n+1} + V_n$$

or

$$V_m - V_n = L_{n+1} - L_n - F \quad \dots\dots\dots(18)$$

MB

To obtain a relation between L_n and L_{n+1} it is necessary to make an enthalpy balance over the feed plate, and to consider what happens when the feed enters the column. If the feed is all in the form of liquid at its boiling point, the reflux L_n overflowing to the plate below will be $L_n + F$. If however the feed is a liquid at a temperature T_f less than the boiling point, some vapour rising from the plate below will condense to provide sufficient heat to bring the feed liquid to the boiling point.

Let H_F be the enthalpy per mol of feed, and H_b the enthalpy of one mol of feed at its boiling point. Then the heat to be supplied to bring feed to the boiling point is $F(H_b - H_F)$ and the number of mols of vapour to be condensed to provide this heat is $F(H_b - H_F)/\lambda$ where λ is the molar latent heat of the vapour.

Then, reflux liquor

$$L_{n+1} = L_n + F + F(H_b - H_F)/\lambda,$$

$$= L_n + F(\lambda + H_b - H_F)/\lambda$$

$$L_{n+1} = L_n + qF \quad \dots\dots\dots(19)$$

$$q = \frac{\lambda + H_{Fs} - H_F}{\lambda}$$

where q = (heat to vaporise 1 mol of feed / (molar latent heat of the feed)

Thus from equation (18)

$$V_m - V_n = qF - F \quad \dots\dots\dots(20)$$

A material balance of the more volatile component over the whole column gives

$$F x_F = D x_D + W x_W \quad \dots\dots\dots(21)$$

Thus combining equations (17), (19) & (20) we get

$$F(q - 1)y_4 = qF x_4 - F x_F$$

or

$$y_4 = \frac{q}{q - 1} x_4 - \frac{x_F}{q - 1} \quad \dots\dots\dots(22)$$

This equation is commonly known as the equation of the q -line. If $x_4 = x_F$, then $y_4 = x_F$. Thus, the point of intersection of the two operating lines lies on the straight line of slope $q/(q - 1)$ passing

(23)

Similarly, taking a material balance for the total streams and for the more volatile component from the bottom to above plate m (indicated by the loop B in Figure 5), and noting that $L_m = L_{m-1}$

and

$$L_m = V_m + W \quad \dots\dots\dots(12)$$

Therefore

$$y_m V_m = L_m x_{m+1} - W x_2 \quad \dots\dots\dots(13)$$

$$y_m = \frac{L_m}{V_m} x_{m+1} - \frac{W}{V_m} x_2 \quad \dots\dots\dots(14)$$

This is similar to equation (11) and gives the corresponding relation between the compositions of the vapour rising to a plate and the liquid on the plate, for the section below the feed plate (stripping section). This equation is the equation of the stripping section or bottom operating line.

5.3.1. Procedure : McCabe-Thiele method

Refer to Figure 6.

- Draw the equilibrium curve, xy diagram (a) and diagonal line, $x = y$ (b)
- Draw the top operating line (c) and bottom operating line (d)
- Starting from where the top operating line cuts the diagonal draw a horizontal line to cut the equilibrium curve at point 1. From point 1 draw a vertical line to cut the top operating line at point 2 (e). Continue in this fashion until where the bottom operating line cuts the diagonal, point 3, remembering to switch to the bottom operating line after the point of intersection of the two operating lines.
- Count the number of triangles drawn. This is equal to the number of theoretical stages required in the column to perform the separation.

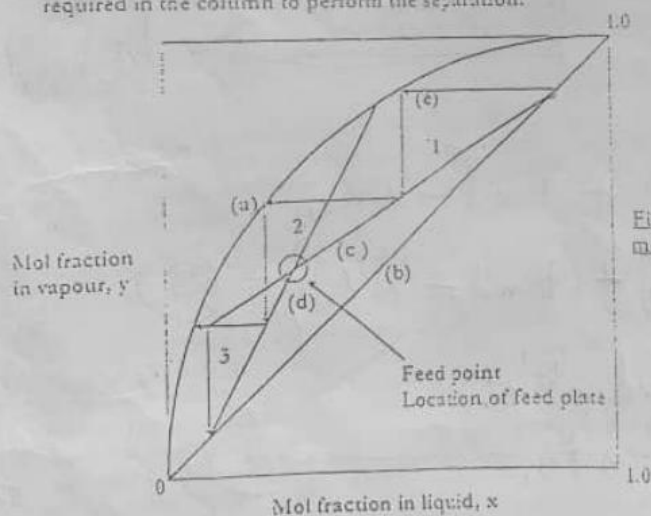


Figure 6 : McCabe- Thiele method

No. of stages = 3

5.3.2. Enrichment in Reboiler (Still) and Condenser

Although the majority of the enrichment occurs in the column proper, some form of enrichment also occurs in the reboiler and condenser. Thus the reboiler and still together represent almost one ideal stage in the distillation column. Thus, the actual number of theoretical plates required is one less than

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through the point (x_f, x_f) . When $y_n = 0$, $x_n = x_f/q$. The line can thus be drawn through two easily determined points. From the definition of q , it follows that the slope of the q -line is governed by the nature of the feed.

(a) Cold feed as liquid	$q > 1$	q line /
(b) Feed at boiling point	$q = 1$	q line
(c) Feed partly vapour	$0 < q < 1$	q line \
(d) Feed saturated vapour	$q = 0$	q line --
(e) Feed superheated vapour	$q < 0$	q line /

Alteration in the slope of the q -line will alter the liquid concentration at which the two operating lines cut each other for a given reflux ratio. This will mean a slight alteration in the number of plates required for the given separation. Whilst the change in the number of plates will usually be rather small, if the feed is cold, there will be an increase in reflux flow below the feed plate, and hence an increased heat consumption from the boiler per mol of distillate.

5.4. The Importance of the Reflux Ratio

5.4.1. Alternate way of drawing operating lines if reflux ratio is known

The ratio L_n/D of the top overflow to the quantity of product is denoted by R , and this enables the equation of the operating line to be expressed in another way, which is often more convenient. Thus, introducing R in equation (11)

$$y_n = \frac{L_n}{L_n + D} x_{n+1} + \frac{D}{L_n + D} x_D$$

$$y_n = \frac{R}{R+1} x_{n+1} + \frac{x_D}{R+1} \quad \dots \dots \dots (23)$$

Thus the equation of the top operating line can be written in terms of the reflux ratio, R . Any change in R will therefore modify the slope of the operating line and will alter the number of plates required for a given separation.

A straight line graph can be simply drawn by joining any 2 points that lie on the line. Therefore to plot the top operating line we need to join any 2 points that satisfy equation (23). If R is known, and x_D is given we can determine the y -intercept of equation (23) i.e. $x_D/(R+1)$, therefore we have one point on the line $[0, x_D/(R+1)]$.

For the second point substitute $x_{n+1} = x_D$ into equation (23), thus $y_n = x_D$ when $x_{n+1} = x_D$, so our 2nd point is $[x_D, x_D]$. By joining these 2 points we obtain the top operating line.

To draw the bottom we again need to join 2 points that satisfy equation (14). Draw the q line, equation (22) and then the top operating line as discussed above, as discussed earlier in this section the point of intersection of the 2 operating lines lies on the q line. We therefore have one point on the bottom operating line i.e. the point where the top operating line and q line intersects. All we need is one more point that satisfies equation (14) to draw the bottom operating line. Let $x_{n+1} = x_f$ in equation (14), then $y_n = x_f$, therefore our 2nd point is $[x_f, x_f]$. By joining these 2 points we can easily draw the bottom operating line.

Hereafter we can step off on the diagram and determine the no. of plates required in the column.

Thus this alternate way of drawing the operating lines is much simpler since there is no need to calculate V_n, L_n, V_m, L_m etc. to use equations (12) & (14).

5.4.2. Influence on the number of plates required

If no product is withdrawn from the still ($D = 0$), the column is said to operate under conditions of total reflux and, as seen from equation (12), the top operating line has its maximum slope of unity,

and coincides with the line $x = y$, the diagonal line. The q line will cut the top operating line on the $x = y$ line and thus the bottom operating line will also coincide with the $x = y$ line. Under these conditions the minimum no. of theoretical stages is required in the column. This can be calculated in the same way that the normal theoretical no. of stages is calculated.

If the reflux ratio is reduced, the slope of the operating line is reduced and more stages are required. Further reduction in R will eventually bring the top operating line to intersect the q line on the equilibrium curve. The reflux ratio cannot be reduced any further and is called the minimum reflux ratio, R_m . This results in an infinite number of stages being required. This arises from the fact that under these conditions the steps become very close together at liquid compositions near x_D , and no enrichment occurs from the feed plate to the plate above. Any small increase in R beyond R_m will give a workable system, though a large number of plates will be required.

If we substitute R_m into equation (23) we get

$$y_n = \frac{R_m}{R_m + 1} x_{n+1} + \frac{x_D}{R_m + 1} \quad \dots \dots \dots (24)$$

Thus if the top operating line is drawn at minimum reflux, its y -intercept is $x_D / (R_m + 1)$.

Two important deductions may be made; first that the minimum number of plates is required for a given separation at conditions of total reflux, and secondly that there is a minimum reflux ratio below which it is impossible to obtain the desired enrichment, however many plates are used.

Exercise

1. An ideal mixture of n -heptane (A) and n -octane (B) consisting of 40 mol% n -heptane and 60 mol% n -octane is to be separated in a distillation column operating at 1 atm total pressure. The feed mixture is at its boiling point. A top product of 90 mol% n -heptane and a bottom of 80 mol% n -octane is required. The reflux ratio, $R = 2.65$.

Data :

$T [^{\circ}\text{C}]$	$P_v^s [\text{mmHg}]$	$P_B^s [\text{mmHg}]$	x_A	y_A
93.4	760	333		
105	940	417		
110	1050	484		
115	1200	561		
120	1350	650		
125.6	1540	760		

- Compute x and y values and draw the equilibrium curve.
- Perform the material balances, obtain the equations of the top and bottom operating lines. Draw the operating lines on the equilibrium curve.
- Determine the theoretical no. of plates required.
- Determine the location of the feed plate.

2.

- Repeat exercise one using the alternate method of drawing the operating lines.
- Determine the minimum theoretical no. of stages required.
- Determine the minimum reflux ratio.

DISTILLATION TUTORIAL

1. A benzene and toluene mixture is separated in a distillation column. The feed stream consists of 40 mol% benzene and 60 mol% toluene and the mixture is fed into the column at boiling point. The reflux ratio is 2. A top product of 90 mol% and a bottom product of 10 mol % benzene is required. Calculate the following using the McCabe-Thiele method.
 - a. The minimum number of stages required.
 - b. The minimum reflux ratio.
 - c. The theoretical number of trays required.
 - d. The location of the feed plate.
2. Repeat question 1 if the feed enters as a saturated vapour and $R = 1.5 \cdot R_{\min}$
3. Repeat question one if $q = 0.75$
4. A mixture of benzene (50 mol %) and toluene (50 mol%) is separated in a distillation column. The distillate contains 95 mol % benzene and the bottom stream 95 mol % toluene. The latent heat of vaporization of the feed is 30 MJ.kmol and the C_p value is 156.4 kJ/kmol.K. Determine the following:
 - a. The q-line if the feed is at 297 K and its boiling point is at a temperature of 353 K
 - b. The minimum reflux ratio.
 - c. The number of trays required if $R = 1.3 \cdot R_{\min}$
 - d. The minimum number of trays.

Equilibrium data for benzene-toluene system:

x	0	0.2	0.4	0.6	0.8	1
y	0	0.43	0.71	0.87	0.96	